

Introduction to the Tidyverse

Import, wrangle, model, and
communicate data

2026-07-30

##



Working with data in R

the tidyverse is a collection of *friendly and consistent* tools for data analysis and visualization.

They live as R packages, each of which does one thing well.

library(tidyverse) will load the core packages:



ggplot2, for data visualisation.

dplyr, for data manipulation.

tidyr, for data tidying.

readr, for data import.

purrr, for functional programming.

tibble, for tibbles, a modern re-imagining of data frames.

stringr, for strings.

forcats, for factors.

lubridate, for dates and times.

This course is hands on!

Each section has an
exercises file:
exercises.qmd

exercises.qmd

exercises.qmd

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ABC

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Render

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C

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↓

Run

🔄

Source

Visual

Outline

1 ---

2 title: "Import Data"

3 format: html

4 ---

5

6 ```{r}

7 #| label: setup

8 library(tidyverse)

9 library(haven)

10 ```

11

12 In this section, we will learn about importing and exporting files from common file formats, including CSV and formats from other statistical software using the readr and haven packages.

13

14 ## readr

15

16 readr supplies several related functions, each designed to read in a specific flat file format.

17

18 Function | Reads

19

readr

Sample data

Importing Data

Your Turn 1

Your Turn 1: Bonus

Tibbles

Missing values

Parsing data types

Your Turn 2

haven: read and write S...

Your Turn 3

Writing data

Your Turn 4

Take Aways

5:1

(Top Level)

Quarto

Code chunks

```
```\{r}  
csv_data <- read_csv(
 "a,b,c,d
 1,2,3,4
 5,6,7,8",
 col_types = ""
)

csv_data
```\
```



Running code chunks

```
```{r}
csv_data <- read_csv(
 "a,b,c,d
1,2,3,4
5,6,7,8",
 col_types = ""
)
```

```
csv_data|
```
```



| a | b | c | d |
|----------|----------|----------|----------|
| <dbl> | <dbl> | <dbl> | <dbl> |
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |

2 rows

Outputting to the console

The screenshot shows the Quarto editor interface with the file 'exercises.qmd' open. The 'Source' tab is active, displaying R code. A context menu is open over the code, with the option 'Chunk Output in Console' selected. The code includes a title '## Parsing data types', a paragraph about the 'readr' package, and a code chunk for reading a CSV file.

```
102 data.  
103 ## Parsing data types  
104  
105 The read functions in readr sometimes it's wrong. For example, the function  
called 'ID' that is a numeric variable. We usually want to treat it as a  
character variable. readr will show you what function to use. You can use the same function  
to read other files.  
106  
107 To do this, add the argument 'col_types' to 'read_csv()' and  
set it equal to a list. readr has several functions that start  
with 'col' that represent data types. We'll go over data and  
object types, including lists, later in the week.  
108  
109 ```{r}  
110 #| eval: false  
111 diabetes <- read_csv("diabetes.csv", col_types = list(id =  
col_character()))  
112 ```
```

The context menu options are:

- Use Visual Editor (⌘F4)
- Preview in Window
- ✓ Preview in Viewer Pane (No Preview)
- ✓ Preview Images and Equations
- ✓ Show Previews Inline
- Chunk Output Inline
- ✓ Chunk Output in Console

The right sidebar shows the 'Outline' pane with the following items:

- readr
- Sample data
- Importing Data
- Your Turn 1
- Your Turn 1: Bonus
- Tibbles
- Missing values
- Parsing data types
- Your Turn 2
- haven: read and write S...
- Your Turn 3
- Writing data
- Your Turn 4
- Take Aways

The status bar at the bottom shows '6:1' and '# Import Data'.

Project contents

```
└─ 01-dplyr_5verbs
  │   └─ cheatsheet_dplyr_5verbs.pdf
  │   └─ diabetes.csv
  │   └─ exercises.qmd
  │   └─ slides.pdf
```